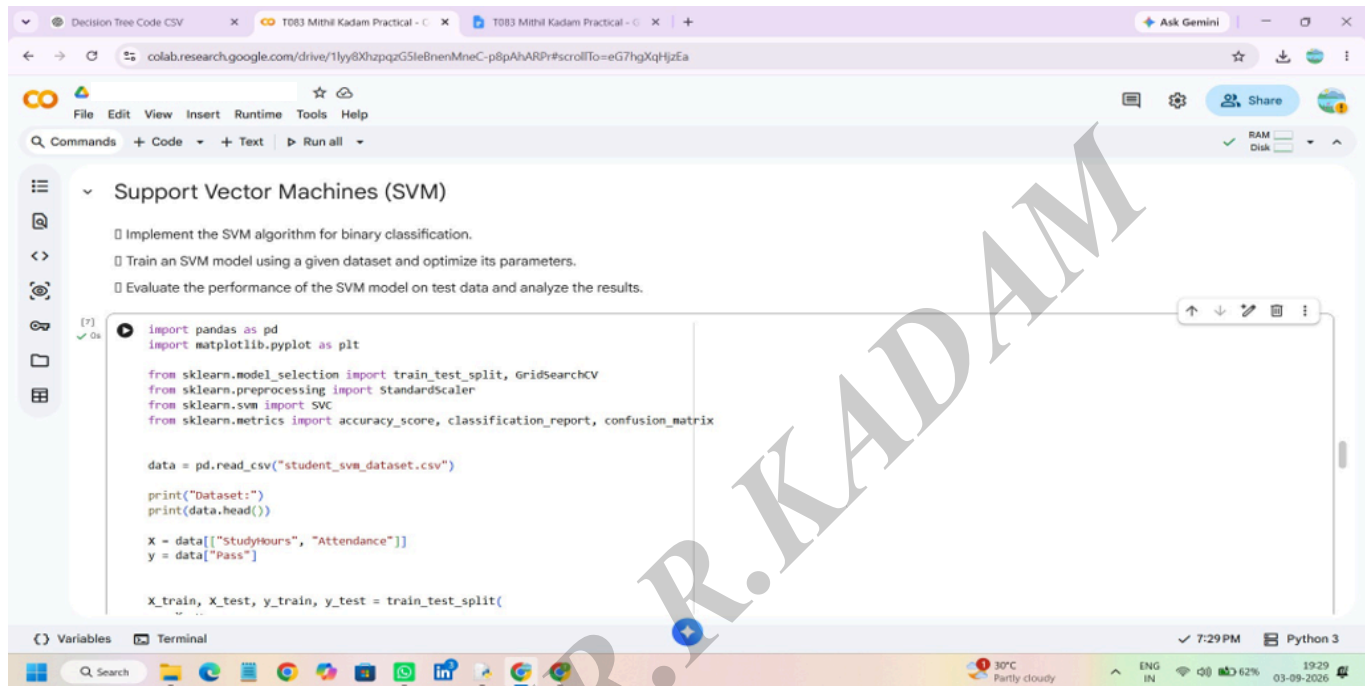


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Aim:- Support Vector Machines (SVM)

Implement the SVM algorithm for binary classification.

Train an SVM model using a given dataset and optimize its parameters. Evaluate the performance of the SVM model on test data and analyze the Results.



The screenshot shows a Google Colab notebook titled "Support Vector Machines (SVM)". The code cell contains the following Python code:

```
import pandas as pd
import matplotlib.pyplot as plt

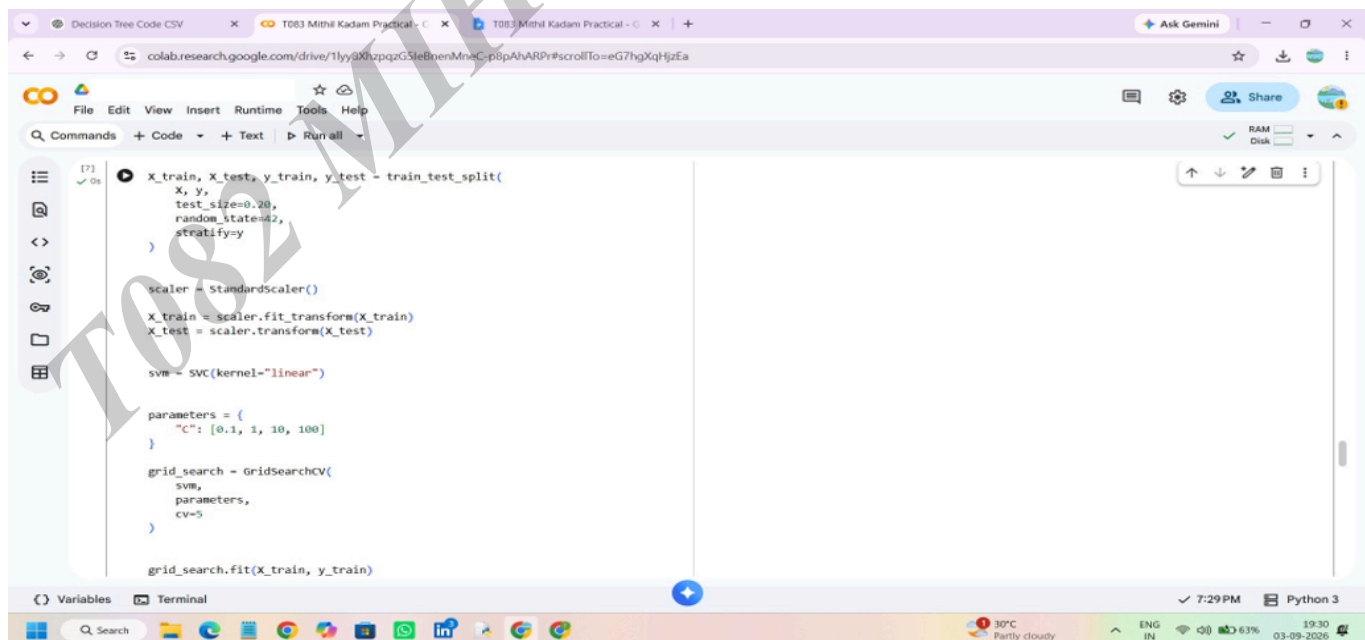
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

data = pd.read_csv("student_svm_dataset.csv")

print("Dataset:")
print(data.head())

X = data[["StudyHours", "Attendance"]]
y = data["Pass"]

X_train, X_test, y_train, y_test = train_test_split(
```



The screenshot shows the continuation of the SVM implementation code in the Google Colab notebook. The code cell contains the following Python code:

```
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

svm = SVC(kernel="linear")

parameters = {
    "C": [0.1, 1, 10, 100]
}

grid_search = GridSearchCV(
    svm,
    parameters,
    cv=5
)

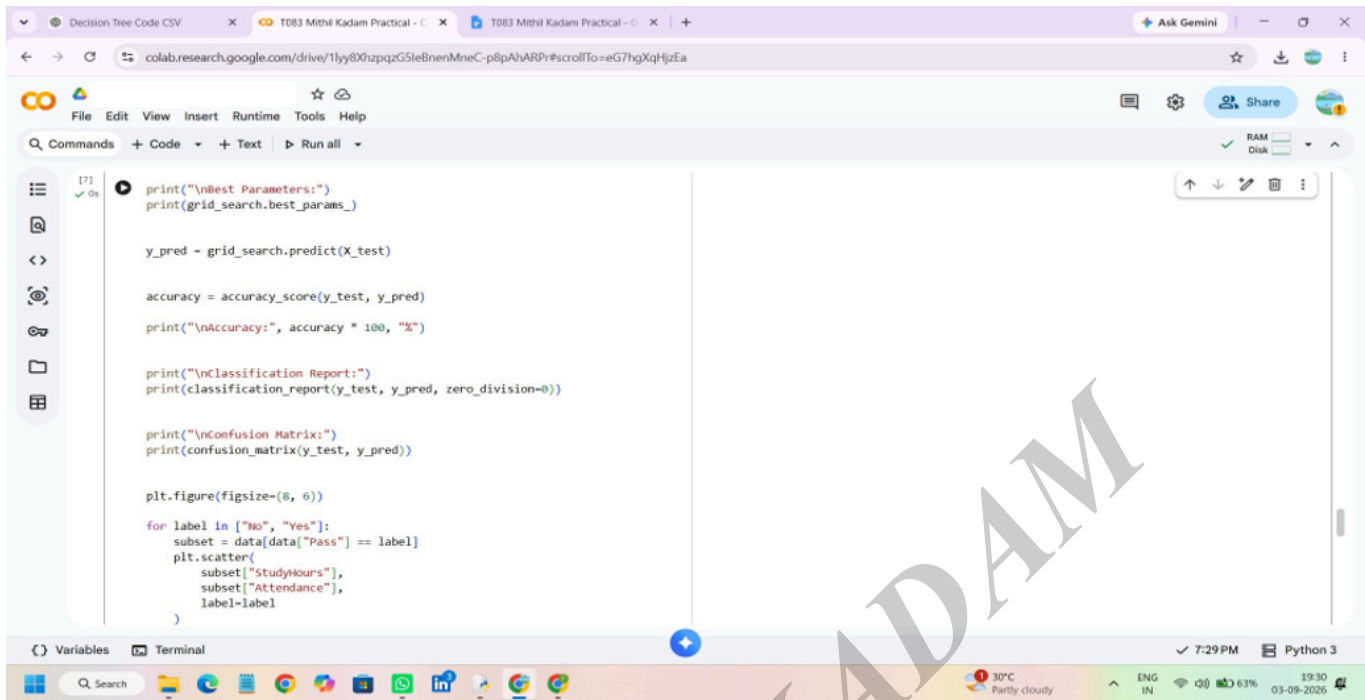
grid_search.fit(X_train, y_train)
```

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This screenshot shows a Google Colab notebook with the following Python code:

```
print("\nBest Parameters:")
print(grid_search.best_params_)

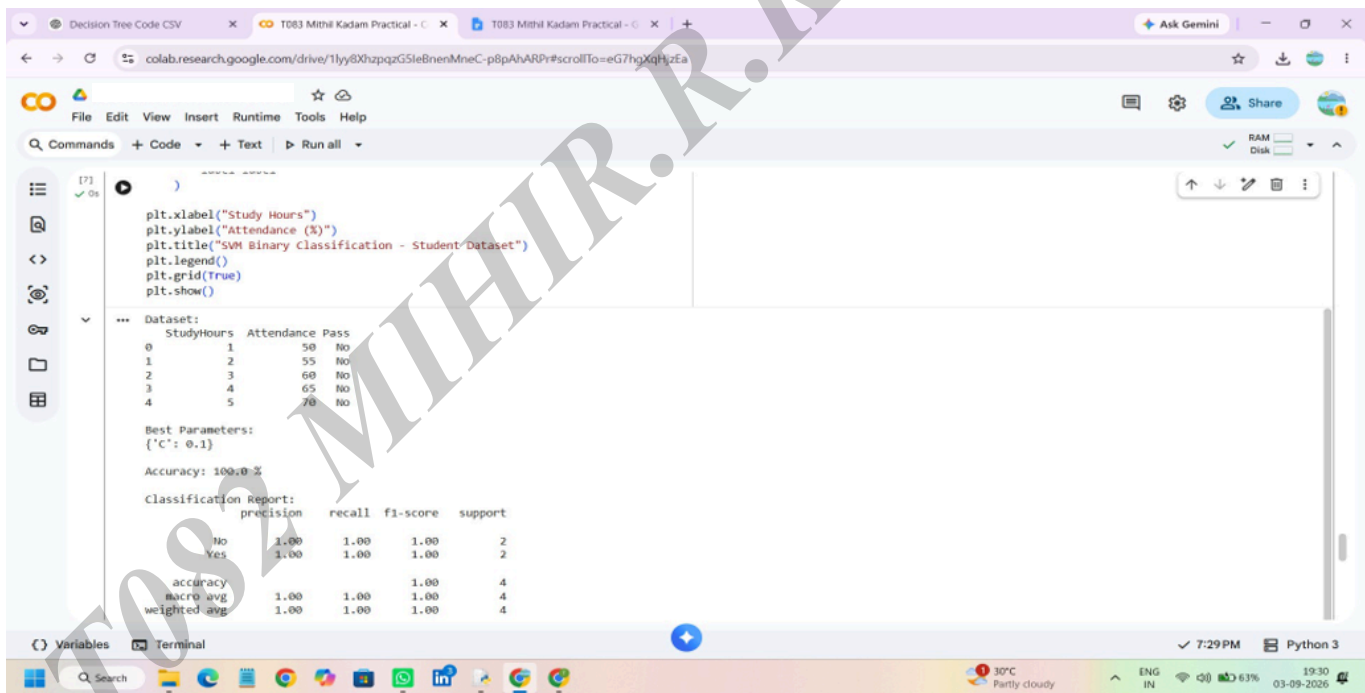
y_pred = grid_search.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred, zero_division=0))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

plt.figure(figsize=(8, 6))
for label in ["No", "Yes"]:
    subset = data[data["Pass"] == label]
    plt.scatter(
        subset["StudyHours"],
        subset["Attendance"],
        label=label
    )
```



This screenshot shows the output of the code from the previous image. The output includes a scatter plot, a dataset table, and classification metrics.

Dataset:

	StudyHours	Attendance	Pass
0	1	50	No
1	2	55	No
2	3	60	No
3	4	65	No
4	5	70	No

Best Parameters:

```
{'C': 0.1}
```

Accuracy: 100.0 %

Classification Report:

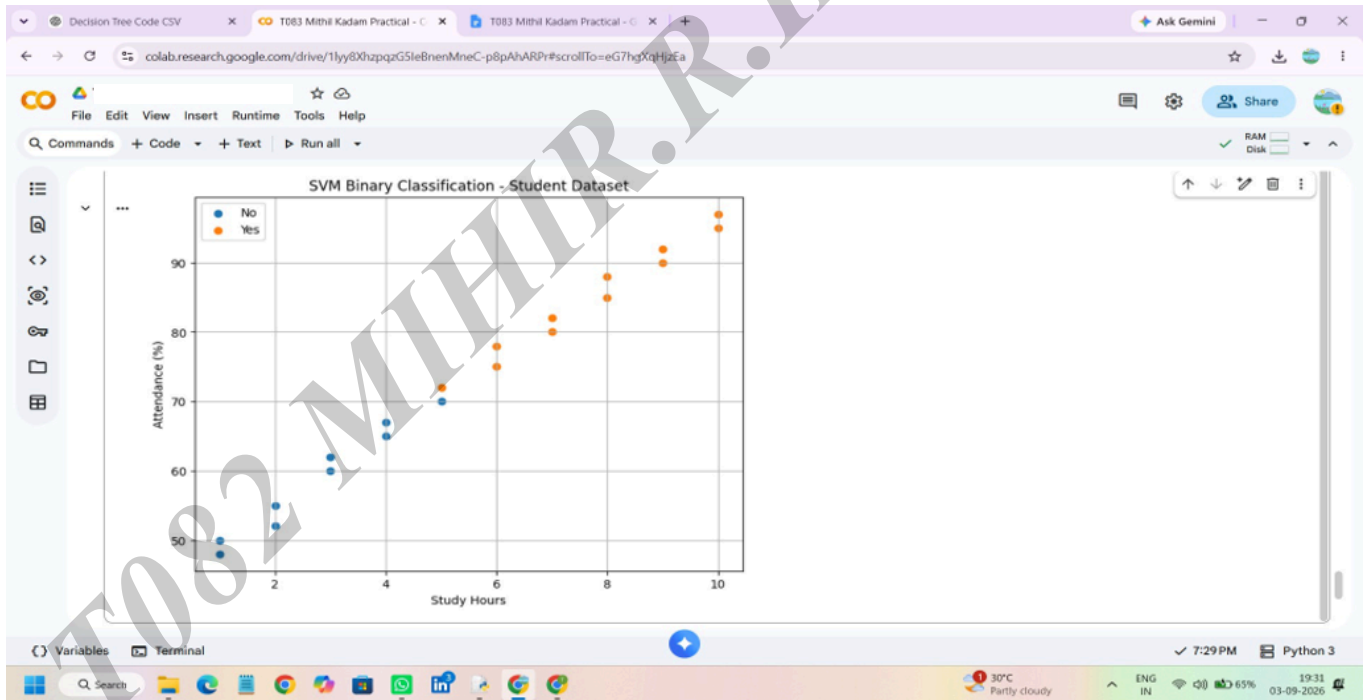
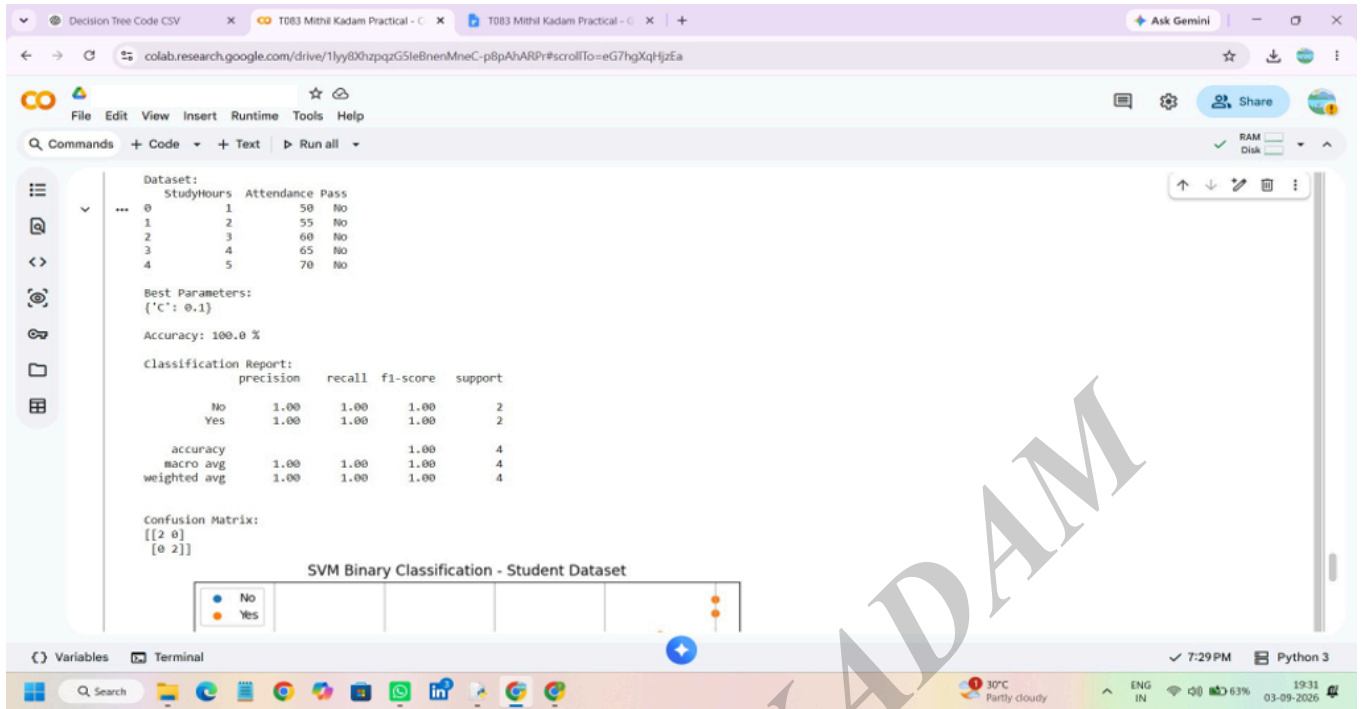
	precision	recall	f1-score	support
No	1.00	1.00	1.00	2
Yes	1.00	1.00	1.00	2
accuracy			1.00	4
macro avg	1.00	1.00	1.00	4
weighted avg	1.00	1.00	1.00	4

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